

Melwin Paul Damiean

Senior Data Engineer

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PROFESSIONAL SUMMARY

- **Senior Data Engineer** with **6+ years of experience** designing, building, and optimizing **data pipelines, cloud data platforms, and machine learning workflows** across **AWS (S3, EMR, Glue, EKS), Azure (Synapse, Databricks), and Snowflake** environments.
- Skilled in **Python (PySpark, Pandas), SQL, and dbt**, with hands-on expertise developing scalable **ETL/ELT frameworks** and **real-time data streaming solutions** using **Apache Kafka, Apache Flink, and Airflow**.
- Experienced in **CI/CD automation** using **Terraform, Docker, and GitHub Actions**, ensuring **high system availability, data quality, and efficient pipeline deployments** across **multi-cloud environments**.
- Adept at leveraging **generative AI, automation, and cloud-native tools** to accelerate **data engineering workflows**, improve **data reliability**, and enable **cross-functional teams** to deliver **analytics and ML-ready datasets**.

TECHNICAL SKILLS

Programming & Scripting: Python, Java, Scala, R, Shell Scripting, SQL (PL/SQL, T-SQL), NoSQL (MongoDB, DynamoDB, Cassandra)

Big Data & Distributed Data Platforms: Apache Spark, Hadoop (HDFS, Hive, MapReduce), Apache Kafka, AWS EMR, Data Lake Architecture

ETL / Data Pipelines & Orchestration: Batch & real-time ETL/ELT Pipelines, Apache Airflow, AWS Glue, dbt Core, Prefect, Pipeline Automation

Data Warehousing & Databases: Snowflake, Amazon Redshift, PostgreSQL, MySQL, Oracle, SQL Server, Data Modelling, Data Governance

Cloud Platforms & Services: AWS (S3, EC2, Lambda, Glue, Athena, EMR, Redshift), Azure (Data Factory, Databricks, Analytics, Azure ML)

DevOps & IaC: GitHub, Jenkins, GitLab CI/CD, Docker, Kubernetes (EKS & AKS), Terraform, Infrastructure Provisioning, Deployment Automation

Machine Learning & MLOps: Scikit-learn, Pandas, NumPy, PySpark, PyTorch, MLflow, Model Deployment & Monitoring, Feature Engineering, NLP, LLM Data Processing, RAG Workflows, Machine Learning Pipelines

BI, Reporting & Visualization: Tableau, Power BI, SSRS, Advanced Excel (Pivot Tables, Formulas), KPI Dashboards

Other Competencies: Agile / Scrum, SDLC, Data Modeling, Data Quality (Great Expectations), Query Optimization, UDFs, Recursive CTEs

EXPERIENCE

MICROSOFT – Senior Data Engineer

Washington, USA | July 2022 – Present

- Designed and implemented a **centralized data transformation framework** using **dbt, Azure Synapse, and advanced SQL** (CTEs, UDFs, query optimization), standardizing **50+ data pipelines** and improving **data integrity by 20%** while reducing development time by **15%**.
- Engineered and maintained **real-time data streaming pipelines** using **Apache Kafka, Flink, Apache Airflow, and Synapse Analytics**, enabling processing of **10M+ daily events** with **sub-150ms latency**.
- Optimized **ETL/ELT performance** in **Azure Databricks (Delta Lake)** by applying **Z-Ordering, Auto Optimize, caching, and adaptive query execution**, achieving **25% faster runtimes** and **12% lower storage costs**.
- Built and automated **machine learning pipelines** in **Python** with **scikit-learn** and **PySpark MLlib**, implementing **feature engineering** and **hyperparameter tuning** to boost **predictive model accuracy by 15%** and cut **false positives by 10%**.
- Leveraged **generative AI** and **LLM-based automation** to accelerate **ETL and ML pipeline creation**, improving development speed by **8%** and ensuring consistent schema and naming conventions.
- Collaborated with **data scientists, product managers, and stakeholders** to translate business requirements into **production-grade data solutions**, ensuring reliable, analytics-ready datasets for **business intelligence** and **machine learning** use cases.

COGNIZANT – Data Engineer

India | July 2017 – December 2020

- Led the **migration of a multi-petabyte enterprise data platform** from **on-premises Hadoop (HDFS, Hive, HBase)** to a **cloud-native AWS architecture (S3, EMR, Glue)**, reducing **total cost of ownership by 25%** and enabling scalable analytics and reporting.
- Designed and managed a **Snowflake data warehouse**, implementing **multi-cluster virtual warehouses, automated data sharing, and role-based access control (RBAC)** to achieve **30% faster query performance** across **10+ business units**.
- Engineered **high-volume ETL/ELT workflows** using **Python (Pandas, PySpark)** and **advanced SQL** to transform, cleanse, and integrate diverse source data, improving **data quality** and cutting **processing time by 15%**.
- Developed **CI/CD automation pipelines** using **GitHub Actions, Terraform, and Docker** deployed on **AWS EKS**, enabling **40% faster deployments** and maintaining **99% system uptime** for production workloads.
- Optimized **Oracle and PostgreSQL queries** through **indexing, execution plan analysis, and partitioning**, enhancing **reporting speed by 25%** for **hundreds of daily users** in **manufacturing analytics** environments.
- Collaborated in an **Agile (Scrum)** environment, participating in **sprint planning, daily stand-ups, and retrospectives**, to ensure **data deliverables aligned with business objectives**, increasing **team velocity by 10%**.

EDUCATION

Master's in Data Science | University at Buffalo, New York, USA.

Bachelor's in Computer Science | Sathyabama University, Chennai, India.